

Chapter 19

Chemical Thermodynamics

1. The first law of thermodynamics can be given as A.

A) $\Delta E = q + w$

B) $\Delta H_{\text{rxn}}^0 = \sum n\Delta H_f^0 (\text{products}) - \sum m\Delta H_f^0 (\text{reactants})$

C) for any spontaneous process, the entropy of the universe increases

D) the entropy of a pure crystalline substance at absolute zero is zero

E) $\Delta S = q_{\text{rev}}/T$ at constant temperature

The first law of thermodynamics: energy cannot be created or destroyed, but can be converted from one to another.

$$\Delta E = q + w$$

2. Which one of the following is always positive when a spontaneous process occurs? C

A) ΔS_{system}

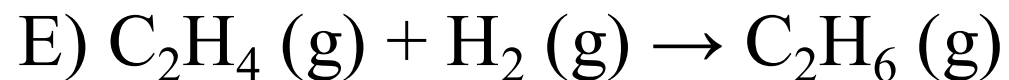
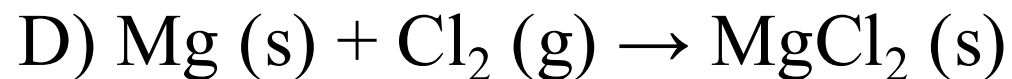
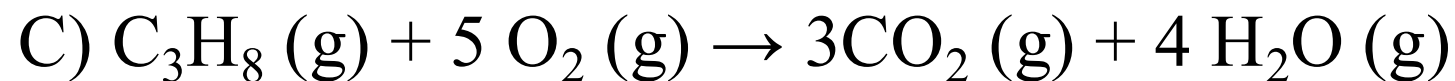
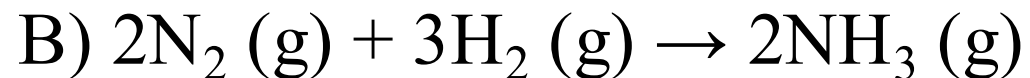
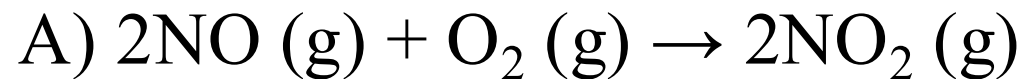
B) $\Delta S_{\text{surroundings}}$

C) $\Delta S_{\text{universe}}$

D) $\Delta H_{\text{universe}}$

E) $\Delta H_{\text{surroundings}}$

3. ΔS is positive for the reaction C.



4. The standard Gibbs free energy of formation of D is zero.

(a) H_2O (l)

(b) Fe (s)

(c) I_2 (s)

A) (a) only

B) (b) only

C) (c) only

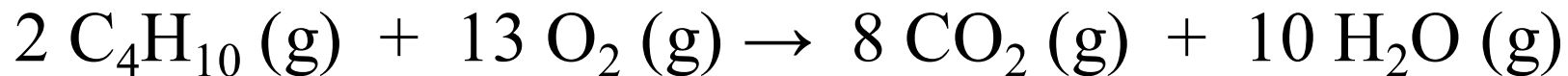
D) (b) and (c)

E) (a), (b), and (c)

Gas
Liquid } 25°C, 1 atm

Element	$\Delta G_f^\circ = 0$ for element in standard state
---------	--

5. For the reaction



ΔH° is -125 kJ/mol and ΔS° is +253 J/K · mol. This reaction is

A.

$$\Delta G^\circ < 0 \quad \forall T$$

- A) spontaneous at all temperatures
- B) spontaneous only at high temperature
- C) spontaneous only at low temperature
- D) nonspontaneous at all temperatures
- E) unable to determine without more information

$$\Delta G = \Delta H - T\Delta S$$

6. Given the following table of thermodynamic data,

Substance	ΔH_f° (kJ/mol)	S° (J/mol $\cdot$ K)
TiCl ₄ (g)	-763.2	354.9
TiCl ₄ (l)	-804.2	221.9

complete the following sentence. The vaporization of TiCl₄ is C.



- A) spontaneous at all temperatures
- B) spontaneous at low temperature and nonspontaneous at high temperature
- C) nonspontaneous at low temperature and spontaneous at high temperature
- D) nonspontaneous at all temperatures
- E) not enough information given to draw a conclusion

7. With thermodynamics, one cannot determine A.

A) the speed of a reaction *kinetics (k)*

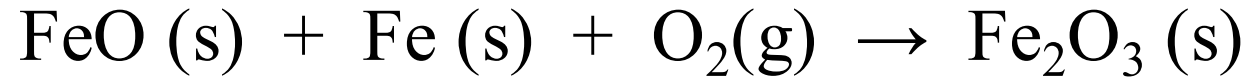
B) the direction of a spontaneous reaction ΔG

C) the extent of a reaction

D) the value of the equilibrium constant $\Delta G = RT \ln K$

E) the temperature at which a reaction will be spontaneous $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

8. Consider the reaction:



Given the following table of thermodynamic data at 298 K:

The value K for the reaction at 25 °C is D.

A) 370

B) 5.9×10^4

C) 3.8×10^{-14}

D) 7.1×10^{85}

E) 8.1×10^{19}

Substance	ΔH_f° (kJ/mol)	S° (J/K • mol)
FeO (s)	-271.9	60.75
Fe (s)	0	27.15
O ₂ (g)	0	205.0
Fe ₂ O ₃ (s)	-822.16	89.96

$-489.79 \text{ kJ/mol} = 831 \text{ J/mol} \cdot \text{K} \cdot 298 \text{ K} \cdot \ln K$
 $K = 7.1 \times 10^{85}$

$\log_{10} e^{197.78} =$
 $197.78 \times \log_{10} e$

9. The normal boiling point of methanol is 64.7 °C and the molar enthalpy of vaporization is 71.8 kJ/mol. The value of ΔS when 1.75 mol of CH₃OH (l) vaporizes at 64.7 °C is C J/K.

A) 0.372

B) 4.24×10^7

C) 372

D) 1.94×10^3

E) 1.94

$$\Delta G^\circ = 0 \quad 337.85$$
$$\rightarrow \Delta S^\circ = \frac{\Delta H^\circ}{T} = \frac{71.8 \text{ kJ/mol} \cdot 1.75 \text{ mol}}{337.85 \text{ K}}$$